

Muhammad Umer Sarfraz Sahi

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PROFESSIONAL PROFILE

Computer Science graduate and Gold Medalist (Ranked 1st in Cohort, CGPA 3.76/4.00) from Khwaja Fareed University of Engineering and Information Technology, with a strong academic foundation in Artificial Intelligence, Deep Learning, and Computer Vision. Demonstrated research capability through an undergraduate thesis applying transfer learning to real-world image classification, complemented by applied experience in Generative AI, Retrieval-Augmented Generation, and full-stack AI system deployment. Motivated by Human-Centered AI and Explainable AI, with emerging interest in Vision-Language Models and Multimodal Machine Learning. Seeking admission to a research-based Master's program to build rigorous methodological expertise and contribute to impactful, reproducible AI research under academic mentorship.

RESEARCH INTERESTS

- Human-Centered Artificial Intelligence
- Computer Vision
- Deep Learning
- Generative AI
- Vision-Language Models (VLMs)
- Multimodal Machine Learning
- Explainable AI (XAI)
- Medical Image Analysis

EDUCATION

B.S. Computer Science

2021 – 2025

Khwaja Fareed University of Engineering & IT (KFUEIT)

Rahim Yar Khan, Pakistan

- CGPA: 3.76 / 4.00 | Gold Medalist – Ranked 1st in Cohort
- Thesis: AI-Based Potato Disease Classification – Transfer Learning (InceptionV3) with a Flask inference API and Next.js frontend for real-time deployment.
- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, AI, Algorithms, Database Systems, Software Engineering.

Higher Secondary School Certificate

2018 – 2020

Punjab Group of Colleges

Khanpur, Pakistan

- Grade: A+

HONORS & AWARDS

- Gold Medalist – Ranked 1st in B.S. Computer Science Cohort, KFUEIT (2025)
- Top Performer Award – HEC Pakistan Generative AI Training Program, Cohort-3 (2026)
- Prime Minister's Youth Laptop Scheme – Government Merit-Based Award (2023)
- Benazir Undergraduate Scholarship (BUSP) – Government Merit Scholarship (2022)
- IELTS Academic – Band 7.5 (Listening 8.5, Reading 7.5, Writing 7.5, Speaking 6.5)

RESEARCH & TECHNICAL PROJECTS

Deep Spud: Potato Disease Classification System

Jul. – Oct. 2024

Final Year Thesis

- Trained a convolutional neural network via InceptionV3 transfer learning to classify multiple potato leaf diseases, applying data augmentation (rotation, zoom, horizontal flip) to improve generalization on a limited agricultural image dataset.
- Engineered a client-server deployment architecture, integrating a Flask REST API for model inference with a React/Next.js frontend, and evaluated performance using accuracy, precision-recall, and confusion matrix analysis.

PK Bazar AI – Agentic Sales Data Analyst

2026

HEC Generative AI Training Hackathon, Cohort-3

- Designed a Retrieval-Augmented Generation (RAG) analytics platform enabling natural-language querying of business data for small enterprises without dedicated data-analysis expertise.
- Collaborated within a team during the HEC Pakistan Generative AI Training Program hackathon (Cohort-3) to design, build, and present the platform under a competitive, time-boxed format.
- Implemented automated ingestion of CSV/XLSX files with trend and seasonality detection; achieved a Top Performer distinction (97.20%).

Brain Tumor Detection via Deep Learning

- Developed a CNN-based MRI classifier using a pre-trained architecture with a full preprocessing pipeline, evaluated using accuracy, precision-recall, and confusion matrix analysis.

AFL AI Assistant – Sports Analytics Agent

2026

Netixsol Internship Capstone Project

- Built a production-oriented conversational AI assistant with a LangGraph multi-node architecture (nine nodes with conditional routing) served via FastAPI, integrating Google Gemini for domain-scoped, tool-augmented chat over pandas-backed sports datasets (7,904 matches; 217,568 player-game rows).
- Trained supporting ML models for match and player-performance prediction — a Gradient Boosting classifier for match winners and Ridge/HistGradientBoosting regressors for player statistics — following a full EDA and feature-engineering pipeline.
- Diagnosed and resolved multi-turn dialogue issues (entity carryover across turns, missing multi-season stat aggregation) to improve retrieval accuracy and conversational reliability ahead of deployment on a live client property.

RealEstate Hub – AI Voice & Chat Agent

2026

LangGraph, Vapi & Deepgram Full-Stack Platform

- Designed a production-grade, stateful LangGraph agent (bilingual English/Roman Urdu) backed by SQLAlchemy and ChromaDB, orchestrating intent detection, RAG-grounded FAQ answering, property recommendation, and appointment booking/rescheduling across nine specialized nodes.
- Integrated a real-time voice interface using Vapi and Deepgram (nova-3) speech-to-text with Google Calendar, CRM, and email tools, and enforced validation invariants preventing double-bookings and hallucinated property listings.
- Implemented full state-transition trace logging (structured JSON and annotated Markdown reports per session) and built a React/Vite frontend to deliver the assistant as a full-stack, deployable product.

PROFESSIONAL EXPERIENCE

Netixsol (via CM IT Internship Program)

Jul. 2026 – Present

AI / Data Science Intern

Faisalabad, Pakistan

- Selected for the competitive CM IT Internship Program and placed at Netixsol to support machine learning workflows end-to-end.
- Support data preprocessing and cleaning workflows for machine learning pipelines, ensuring dataset quality and consistency.
- Assist in developing and evaluating machine learning models in Python, contributing to feature analysis and performance benchmarking.

The Journal Post

Mar. 2025 – Jul. 2026

Web Developer

Lahore, Pakistan

- Architected and optimized high-performance web apps using React.js and Next.js, improving Core Web Vitals via server-side rendering and SEO strategies.
- Developed reusable, accessible UI component libraries and integrated secure REST APIs, applying modern authentication practices and WCAG accessibility compliance.

Freelance Virtual Assistant – Amazon FBA Store Management

Sep. 2024 – Feb. 2025

Self-Employed

Remote

- Managed Amazon FBA store operations for an international e-commerce client, running PPC campaigns, optimizing listings via keyword/competitor research, and handling customer service to maintain account health.

The Journal Post

Jun. – Aug. 2024

Web Developer Intern

Lahore, Pakistan

- Implemented SSR and static generation in Next.js to improve performance and SEO rankings, and managed complex app state using Redux and Context API.

TECHNICAL SKILLS

- Programming: Python, JavaScript (ES6+), TypeScript, SQL
- AI / ML: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, CNNs, Transfer Learning, RAG, LLM Integration
- Web Development: React.js, Next.js, Flask, Node.js, REST APIs, Redux
- Databases: PostgreSQL (Supabase), MySQL
- Developer Tools: Git, Docker, Jupyter Notebook, Google Colab, Figma

CERTIFICATIONS

- Generative AI Application Developer Certificate – ULEF USA, Pak Angels, iCodeGuru, ASPIRE Pakistan, HEC Pakistan (2026)
- Google Digital Marketing & E-Commerce Professional Certificate – Google, via Coursera (Dec 2025)
- Google AI Essentials Specialization Certificate – Google, via Coursera (Sep 2026)
- WORC Completion Certificate – Enablers (Nov 2024)

LANGUAGES

English (Proficient – IELTS Academic Band 7.5) | Urdu (Native) | Punjabi (Native)